



This is a course in single-variable calculus. This course covers limits, continuity, derivatives and their applications, definite and indefinite integrals, infinite sequences and series, plane curves, polar coordinates, and basic differential equations. This course is approved by the College Board as an AP Calculus BC class, however, “AP exam preparation” is not the main focus of the course.

Calculus is specifically designed for high-performing students and draws material from many programs for top high school students in the country. Our philosophy is that students develop more by learning to solve problems they haven’t seen before, as opposed to offering repeated drills that students can memorize their way through. In this way, our classes are structured much more like courses at top-tier colleges.

In comparison to many high school calculus courses, there are two main things we hope to accomplish:

- Students will gain a fundamental understanding of single-variable calculus, beyond the level of rote calculation.
- Students will learn how to apply calculus techniques to solve difficult problems.

However, our course is not “rigorous” in the sense that a college-level Real Analysis course would be. Although we will try to gain a deeper understanding of many calculus concepts, we will not commit to rigorously proving every result.

Book: *Calculus*, by David Patrick

Time Commitment: This 36-week course includes 105 minutes in-class and 1-3 hours of homework each week, corresponding to a full year course.

Content:

Week	Topic
1	Interpretation of the Derivative
2	Trig Functions, Logs, Exponentials
3	Limits
4	Limits: Continuity & Infinity
5	Definition of the Derivative
6	Linear Approximation & MVT
7	Basic Derivative Computations
8	The Chain Rule
9	Implicit Differentiation
10	Related Rates
11	Optimization
12	Midterm 1
13	Area Under a Curve
14	Fundamental Theorem of Calculus
15	Numerical Integration
16	Simple Substitution
17	Integration by Parts
18	Trig Substitution

Week	Topic
19	Partial Fractions
20	Volumes
21	Arc Length & Surface Area
22	Improper Integrals I
23	Improper Integrals II
24	Midterm 2
25	L’Hopital’s Rule
26	Sequences and Series
27	Series Convergence I
28	Series Convergence II
29	Power Series
30	Taylor Polynomials
31	Taylor Series
32	Parametric Curves
33	Polar Coordinates
34	Differential Equations I
35	Differential Equations II
36	Final Exam

Common Core State Standards:

Domain	Subdomain	Standards
Functions	Interpreting Functions	4, 6, 7a
	Trigonometric Functions	5, 6, 8, 9

Sample Problems:

- Let f and g be functions with domain $\mathbb{R}$. Suppose

$$\lim_{x \rightarrow a} f(x) = b \text{ and } \lim_{x \rightarrow b} g(x) = c.$$

Prove or disprove that we must have $\lim_{x \rightarrow a} (g \circ f)(x) = c$. (If true, explain why, with a rigorous proof if possible, if false, give an example.)

- Sam wishes to cross a circular lake with diameter 1 km. He can row across the water at a rate of 4 km/hour, or he can walk along the shore (carrying the rowboat) at a rate of 6 km/hour. What is the minimum amount of time necessary to cross the lake?
- A 4-dimensional unit sphere is the set of 4-tuples (w, x, y, z) that satisfy $w^2 + x^2 + y^2 + z^2 = 1$. Compute its volume.
- Suppose a, b, x, y are all positive and $a + b = 1$. Compute $\lim_{t \rightarrow 0^+} (ax^t + by^t)^{\frac{1}{t}}$.
- The Fibonacci numbers are defined as $a_0 = a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for all $n \geq 2$. Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n = 1 + x + 2x^2 + 3x^3 + 5x^4 + 8x^5 + \cdots.$$

- (a) Prove that the radius of convergence of f is at least $\frac{1}{2}$.
- (b) Prove that if $|x| < \frac{1}{2}$, then $f(x) = \frac{1}{1 - x - x^2}$.